



Network structure and the journey to work: An intra-metropolitan analysis

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ABSTRACT

This research quantifies the variation of network structure within the Minneapolis - St. Paul metropolitan area and relates it to average travel time to work for each Minor Civil Division (MCD) in the metro area. The variation of these measures within the metropolitan area is analyzed spatially. The measures of network structure are then related to observed travel. Better connected networks have lower average travel times, all else equal. The results corroborate a relation between network structure and travel and point to the importance of understanding the underlying street network structure.

1. Introduction

Recent advances in spatial analysis and modeling techniques combined with significant improvements in computing power have resulted in a revival of interest in understanding complex networks (Gastner and Newman, 2006; Strogatz, 2001). The current interest in network analysis ranges from empirical studies of specific network systems to the development of new models and techniques to analyze and interpret complex systems (Clauset et al., 2007). A purely topological approach to complex networks works well for virtual networks such as the world wide web but need not necessarily work for physical networks that have additional geographic properties (Kalapala et al., 2006). Hence, the traditional focus on network topology needs to be complemented with consideration of the pivotal role of geography or space (Boccaletti et al., 2006). For example, the nodal degree (number of edges connected to a node) is limited by the available physical space, specifically on planar networks like urban street networks. This in turn influences the structural properties of the network. A complete understanding of transport networks thus needs to include spatial aspects such as location, distance, and geography (Gorman and Kulkarni, 2004).

Transport networks that connect points in geographic space easily lend themselves to spatial analysis and the interest in understanding their structural attributes appeals to researchers from various backgrounds. For example, physicists have now started looking at the geographical and spatial aspects of transport networks to understand and reproduce their qualitative features (Gastner and Newman, 2006). Transport networks are also one of the few complex networks that are resilient and preserve their geometric structure as they evolve over time (Mohajeri and Gudmundsson, 2014). Researchers have therefore used transport networks to understand the evolution trends in complex networks, the connection patterns in real-world networks, and the transferability of small-world concepts to real-world networks (Derrible, 2012; Guimerà et al., 2006; Latora and Marchiori, 2002).

The structure of transport networks, with investigations starting from the 1960s (Garrison and Marble, 1961; Haggett and Chorley, 1969; Taaffe and Gauthier, 1973), holds specific interest to geographers due to its influence on human patterns and

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economic development (Wei and Yao, 2014). Graph theory-based measures of network structure characterize the underlying structure of transport (railways and roadways) networks and relate it to regional economic characteristics (Kansky, 1963).

The interest in analyzing the structure of street networks to measure the spatial properties of cities was further propelled by the use of the ‘space syntax’ technique, originally proposed by Hillier and Hanson (Barthélemy, 2011). Space syntax originated in the early seventies and refers to a family of techniques used to represent, quantify and interpret the spatial configurations in buildings and settlements (Hillier and Hanson, 1997). Space is viewed as an intrinsic aspect of human activity and the basic theory of ‘space syntax’ is that social relations and processes express themselves primarily through spatial configurations (Hillier et al., 1987).

Both graph theory-based measures and space syntax approaches to understanding the spatial structure of networks were hampered by limited data, computational resources (Gastner and Newman, 2006) and the difficulty in estimation and interpretation (Marshall and Garrick, 2012). In recent years, network analysis has come back into the limelight and pervades different fields ranging from neurobiology to statistical physics (Strogatz, 2001).

Understanding the spatial and structural patterns of street network has numerous applications in urban planning, transport, geovisualization and location-based services (LBS) (Yang et al., 2009). The importance of understanding street network structure in the disciplines of architecture, road construction and urban management have been highlighted as well (Marshall, 2005). Graph theory based network topology approaches have been used in studying the vulnerability and resiliency of transport systems (Mattsson and Jenelius, 2015). In other applications, researchers have also looked into how street network structure and design influences non-motorized traffic safety (Zhang et al., 2015), non-motorized travel behavior (Schoner and Levinson, 2014), physical activity (Koohsari et al., 2016), retail activity (Omer and Goldblatt, 2016) to name a few.

The research interest in spatial network analysis is now being extended to incorporate new ‘big data’ sources (e.g. transit smart card data, GPS-enabled taxi data, social network data, etc.) to better understand human mobility, interactions and urban transformations (Liu et al., 2015; Saberi et al., 2016; Zhong et al., 2014). In the travel behavioral field, current efforts have focused on relating network structure to transport system performance (Jiang et al., 2009; Jenelius, 2009; Derrible, 2010). The spatial configurations of street networks have been used to understand traffic distribution and system efficiencies (Zadeh and Rajabi, 2013) and evaluate the impacts of space removal in urban settings (Ortigosa and Menendez, 2014).

This paper extends this research interest in network structure and travel. The variation of network structure between metropolitan areas and its influence on transport system performance was confirmed by recent research (Parthasarathi, 2014). This paper examines the variation of network structure within a metropolitan area. This variation in network structure is then related to observed travel using spatial and econometric analyses. The average travel time to work is used as a measure of observed travel.

This paper is organized as follows: the next sections provide a detailed explanation of the modeling methodology, namely the data, model formulation and the estimation of measures of network structure. The spatial and regression analyses are presented in the following sections. The paper concludes with a discussion of the results and its implications.

2. Data

2.1. Minor civil divisions

The data for this analysis come from the Minor Civil Divisions (MCD) within the Minneapolis-St. Paul (Twin Cities) metropolitan area. MCDs are defined as primary divisions of a county, and generally provide some governmental services. The US Census Bureau recognizes MCDs in 28 states and Puerto Rico (U.S. Census Bureau, 2011).

The MCD dataset was downloaded from the Regional MetroGIS database in GIS shape file format (MetroGIS DataFinder, 2011). The downloaded MCD dataset represents municipal boundaries and includes cities, townships and unorganized territories within the Twin Cities metropolitan area. The Twin Cities metropolitan area refers to the seven counties of Anoka, Carver, Dakota, Hennepin, Ramsey, Scott and Washington and includes the cities of Minneapolis and Saint Paul. There are 189 MCDs within this metropolitan area.

The area of the MCD (km²) was estimated using ArcGIS for each MCD in the dataset. This is used as a control variable in the regression models estimated later in this paper.

2.2. Street networks

The street network for this analysis was extracted from the US Census TIGER/Line files. The TIGER/line files are spatial extracts from the US Census Bureau’s Master Address File (MAF)/Topologically Integrated Geographic Encoding and Referencing (TIGER) database (MAF/TIGER). This database consists of select cartographic and geographic information and provides information on various features such as roads, railroads, rivers, as wells as legal and statistical geographic areas (U.S. Census Bureau, 2008). The extracted Twin Cities network was cleaned to include just the road features based on the Feature Class Codes (FCC) for the line segments provided in the Census TIGER/Line files.

For the purposes of this current analysis, the extracted Twin Cities street network was overlaid on the MCD shape file to extract the street network for each MCD. These extracted street networks within each MCD were cleaned to ensure that the network contained only the relevant links and intersection or junction nodes. These networks were then used to estimate measures of network structure within each MCD, which are elaborated in a later section of this paper.

2.3. Census data

The downloaded MCD dataset was supplemented with the year 2010 travel and socio-demographic data from the US Census Bureau (U.S. Census Bureau, 2010). The variables of interest for each MCD in the Census data include the average travel time to work and the median household income.

2.4. Person weighted accessibility

Additional data used in this paper include multimodal accessibility estimates for each MCD obtained at the Traffic Analysis Zone (TAZ) level, estimated as part of a research project at the University of Minnesota (Anderson et al., 2013). The study used accessibility estimates as a performance measure in evaluating potential land use and transport scenarios for the Twin Cities metropolitan area.

There are various forms of accessibility measures and the “cumulative opportunity” measure, a commonly used accessibility measure, was used in this study. This measure evaluates accessibility as the number of opportunities available within a specific time or distance of a defined origin or reference point (Handy, 2002; Tasic et al., 2017). The cumulative opportunity accessibility measure is typically defined as follows (El-Geneidy and Levinson, 2006; Levinson et al., 2010):

$$A_i = \sum_j O_j f(C_{ij}) \quad (1)$$

$$f(C_{ij}) = \begin{cases} 1 & \text{if } C_{ij} \leq T_c \\ 0 & \text{if } C_{ij} > T_c \end{cases} \quad (2)$$

where

- A_i = for the job accessibility of origin i ,
- O_j = for the number opportunities (e.g. jobs) at destination j ,
- C_{ij} = for the costs between origin i and destination j
- T = the corresponding cost threshold.

Cumulative accessibility to population, employment, and labor within a specified time threshold, ranging from 10 min to 60 min, were estimated for each TAZ in the Twin Cities metropolitan area.

In addition to the cumulative accessibility measure, the study also provided a person-weighted accessibility measure. The person-weighted accessibility for each zone, A , was estimated by multiplying the cumulative accessibility of the zone within a certain time threshold (e.g. 30 min) by a weight, W_i and dividing the product by the sum of weights, as shown below in Eq. (3). Weight, W_i , is the population or number of workers or jobs in that zone experiencing that level of accessibility.

$$A = \frac{\sum_{i=1}^I A_i * W_i}{\sum_{i=1}^I W_i} \quad (3)$$

The use of a person-weighted accessibility measure allows us to summarize the TAZ level accessibility data to a larger area such as states, metropolitan planning areas and MCDs, as used in this analysis. “The person-weighted approach allows the summary metric to reflect the distribution of residents within the area. For example, a person-weighted accessibility value of 134,173 indicates that a typical resident within the area can reach 134,173 jobs”. (Accessibility Observatory, 2016)

The estimated person weighted accessibility to jobs within the specified time threshold using auto and transit for each MCD is used in this analysis.

3. Estimation of network measures

The following measures of network structure were estimated using the street network structure for each MCD. The measures of network structure were designed to capture different aspects of the underlying street network and fall into the following broad categories:

- Hierarchy - refers to the differentiation that exists in street networks.
- Topology - identifies the connectivity and the connection patterns that exists in street networks.
- Scale - captures the intensity of the street network within a specified area.

Table 1 summarizes the estimation of the above network measures, compiled from various sources. To further elaborate on the estimated network measures, Fig. 1(a)–(f) provides a visual representation of select network measures using frequency distribution plots. The correlation between the various estimated measures is presented in Table 2. It is important to point out that many of the topology measures identified here are graph-theory based measures of network structure.

Table 1
Estimation of network measures.

Network Measures (Unit)	Equation	Notes	Reference
Entropy	$H(X) = -\sum_{i=1}^I p_i \log_2(p_i)$	I = Number of different functional classifications of roads, p_i = Proportion of links in each functional classification, i , in MCD ^a , $i = 1$ to I	Xie and Levinson (2007)
Percent freeways	$F = \frac{L_{fm}}{L_{sm}} * 100$	L_{fm} = Length (km) of the freeways within the minor civil division, L_{sm} = Length (km) of the street network within the minor civil division	
Alpha ^b	$\alpha = \frac{e-v+1}{2v-5}$	e = Number of links or edges in the street network within the minor civil division, v = Number of nodes or vertices in the street network within the minor civil division, $e-v+1$ = The numerator is also referred to as the cyclomatic number, u , commonly used in graph theory	Kansky (1963)
Arterial Treeness ^c	$\phi_{tree} = \frac{L_{tm}}{L_{sm}}$	L_{tm} = Length (km) of street segments belonging to a branch or tree network within the minor civil division	Haggett and Chorley (1969), Xie and Levinson (2007)
Average Circuity	$C_t = \frac{D_{nm}}{D_{em}}$	D_{nm} = Sum of the network distance (km) between all OD pairs in the minor civil division, D_{em} = Sum of the euclidean distance (km) between all OD pairs in the minor civil division	Ballou et al. (2002), Levinson and El-Geneidy (2009)
Eta (km)	$\eta = \frac{L_{sm}}{e}$		Kansky (1963)
Theta (km)	$\theta = \frac{T_m}{v}$	T_m = Total traffic within the minor civil division, measured here as Daily Vehicle km Traveled (DVKT)	Kansky (1963)
Street density (1/km)	$\rho_{lm} = \frac{L_{sm}}{A_m}$	A_m = Area (km ²) of the minor civil division	

^a Functional classifications are identified by FCC codes in the Census TIGER/line file.

^b $\beta = \frac{e}{v}$ and $\gamma = \frac{e}{3v-6}$ measures were estimated as well. Only α is considered in the analysis due to the extremely high correlation (almost 1.0) between the three measures. The extremely high correlation confirms the three measures are one and the same. Refer to Levinson (2012) for additional explanation.

^c Open source software was used to classify each segment in the street network as belonging to a tree network or a circuit network. Code was developed by Feng Xie, Metropolitan Washington Council of Governments (MWCOC) and can be downloaded from <https://conservancy.umn.edu/handle/11299/181333>.

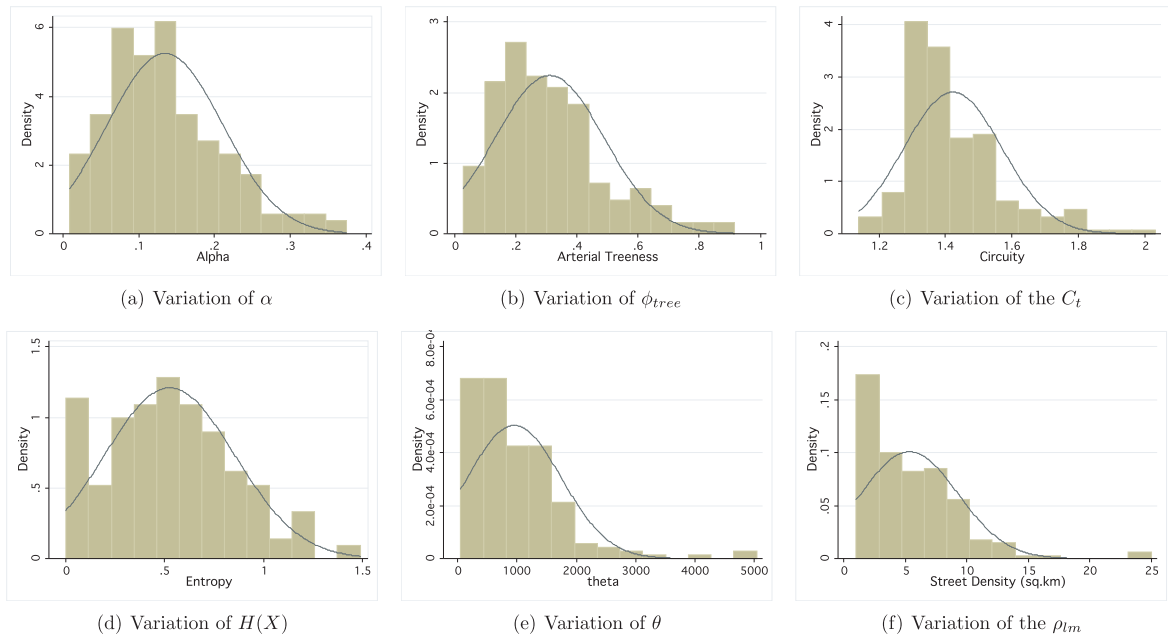


Fig. 1. Distribution of estimated network measures.

Table 2

Correlation of estimated measures of network structure within the Minor Civil Division.

	Entropy, $H(X)$	Percent freeways, F	Alpha, α	Circuitry, C_t	Arterial Treeness, ϕ_{tree}	Eta, η	Theta, θ	Street density, ρ_{lm}
Entropy, $H(X)$	1							
Percent freeways, F	0.303	1						
Alpha, α	−0.097	0.177	1					
Circuitry, C_t	0.043	−0.013	−0.503	1				
Arterial Treeness, ϕ_{tree}	−0.025	−0.197	−0.728	0.340	1			
Eta, η	0.160	−0.194	−0.389	0.156	0.080	1		
Theta, θ	0.420	0.523	0.107	−0.021	−0.197	0.187	1	
Street density, ρ_{lm}	−0.126	0.242	0.692	−0.288	−0.410	−0.630	−0.103	1

Bold - indicates significance at 95% confidence level.

4. Spatial analysis

The objective of this paper was to analyze the variation of network structure within a metropolitan area and relate this variation to observed travel. The first step in this analysis was to look at the spatial variation of the network structure across the MCDs, as presented in Fig. 2(a)–(e). To better understand the spatial variation, the region's limited access highway network is identified in each of the figures, as are the core cities of Minneapolis and St. Paul.

**Fig. 2.** Variation of estimated network measures by Minor Civil Division.

Consider the variation of the estimated α index, shown in Fig. 2(a). As pointed out previously, the α index measures the degree of connectivity within the street network. The α index varies between 0 and 1 and a higher value of the α index indicates higher connectivity in the network. From Fig. 2(a), we can see a clear pattern emerge across the metropolitan area. The α index is highest in the center of the Twin Cities metropolitan area within the beltway and the value progressively decreases as we move out towards the edge. The street networks in the central part of the metropolitan area (including the cities of Minneapolis and St. Paul), have a typical, though imperfect grid-like structure and therefore show a higher α index. The outer part of the metropolitan area typically has a meandering dendritic network which corresponds to the lower value of the α index.

The spatial variation of the ϕ_{tree} measure is presented in Fig. 2(b). The variation of ϕ_{tree} shows that the grid networks in the central part of the metropolitan area have a lower ϕ_{tree} value. The outer edge of the metropolitan area show higher ϕ_{tree} values.

The corresponding variation of C_t measure is shown in Fig. 2(c). Like the ϕ_{tree} measure, the lowest C_t value is seen in the central cities of Minneapolis and St. Paul and the highest C_t value is seen in the MCDs on the outer edge of the metropolitan area. This variation in ϕ_{tree} and C_t measures confirm that the grid network, with a lower proportion of tree-like segments, provide more direct routes for travelers, resulting in lower C_t values. On the other hand, suburban networks with a higher proportion of tree-like segments, provide less direct routes for travelers, resulting in higher C_t values.

Now consider the spatial variation of the η index presented in Fig. 2(d). The η index measures the average link or edge length. Fig. 2(d) shows that the η value progressively increases from the central part of the metropolitan area towards the outer edge of the metropolitan area. The central cities of Minneapolis and St. Paul have the lowest η value. This shows that the average edge length is lower for grid-like street networks compared to tree-like street networks.

The variation in the η index corresponds to the variation in ρ_{lm} , presented in Fig. 2(f). The ρ_{lm} variable captures the intensity of the street network within a given area and is highest in the central part of the metropolitan area and progressively decreases towards the outer edge of the metropolitan area.

The estimated network measures presented in Fig. 2(a)–(e) show the spatial variation across MCDs in the metropolitan area. The spatial analysis confirms the difference in network structure between the grid-like street networks seen in the central portion of the Twin Cities metropolitan area and the tree-like street networks seen in the outer edges of the metropolitan area. The estimated network measures help quantify the differences in street network structure across the metropolitan area.

The spatial analysis also point to a relationship between certain network measures. To expand on this, scatterplots of select network measures were analyzed and are presented in Fig. 3(a)–(e). Consider the relationship between the α index and C_t measure presented in Fig. 3(b). This scatterplot shows that the circuitry in the network decreases with increased connectivity as measured by the α index. Similarly the treeness in the network decreases with increased connectivity, as seen in the scatterplots of Fig. 3(c). Another scatterplot comparison between the θ index and the $H(X)$ index is presented in Fig. 3(f). The scatterplot shows a positive relationship between the two variables in that the traffic load of the network, measured by θ increases with increasing $H(X)$. Please note that the scatterplots presented here highlight specific relationships between select network variables and do not cover all variables estimated in Table 1.

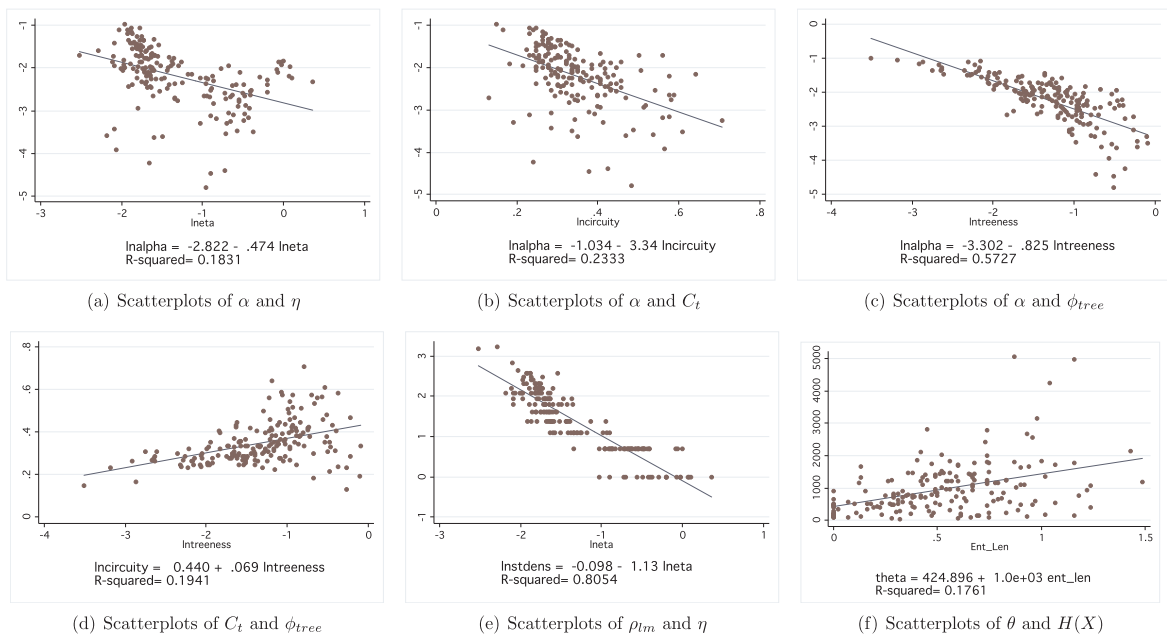


Fig. 3. Scatterplots of select network measures.

5. Hypotheses

Based on the above spatial and scatterplot analyses, a set of core hypotheses were formulated for the various network measures listed in Table 1. These hypotheses are listed below:

- H1: Aspects of network structure that increase the connectivity of the street network within an MCD (operational variables - α) will decrease the average travel time to work.
- H2: Aspects of network structure that increase the network travel distance (operational variables - ϕ_{tree} , C_t) between fixed origins and destinations within an MCD will increase the average travel time to work.
- H3: Aspects of network structure which increase the differentiation in the street network (operational variables: $H(X)$, F) within an MCD will reduce the average travel time to work. These measures capture the presence of higher hierarchy links such as interstates, which typically have higher speeds and reduce travel time.
- H4: An increase in the MCD's person-weighted accessibility to jobs within a time threshold (30 min here) by auto or transit ($A_{A,30}$, $A_{T,30}$) lowers the need for longer travel and reduces the average travel time to work.

It is important to note that while the above set of the hypotheses could apply to a broad set of network measures, the estimated models include only select network measures. Each of the independent network variables were included sequentially in the models based on the estimated correlation between the network variables and the statistical significance of each of the variables.

6. Categorical analysis

The analyses in the previous section shows the spatial variation of estimated network measures within the metropolitan area and also highlight specific relationships between select network measures. The analysis in this section relates the estimated network measures within the MCD to observed travel using statistical regression models. The 2010 average travel time to work in minutes from the US Census Bureau is used as a measure of observed travel for each MCD. Before undertaking statistical models, a simple categorical analysis of the relation between the estimated network measures and observed travel was conducted.

The grouping of the MCDs into the above four categories was based on a review of the observed travel time to work, obtained from the 2010 Census. The minimum observed travel time to work was 18.2 min and the maximum observed travel time was 52.9 min. The grouping of the MCDs into the above listed categories was based on the associated frequency distribution of the average travel time to work.

1. (0,20] min - 5%
2. (20,25] min - 45%
3. (25,30] min - 35%
4. (30, ∞) min - 13%

The mean value of the network measures, listed in Table 1, was calculated for each of the time categories, and tabulated in Table 3. For example, consider the mean value of $H(X)$, which measures the differentiation in the street network. The tabulation shows that the entropy, $H(X)$, is highest for category 1 and lowest for category 4. As explained in Table 1, a higher entropy value indicates the presence of higher functional classification links in the network such as arterials compared to local streets. The higher functional classification links are typically faster and result in lower travel times.

Similarly the treeness variable, ϕ_{tree} , which measures the presence of tree-like segments, is lowest for category 1 and is the highest for category 4. A tree-like network provides minimum connections and hence longer distances between the fixed origins and destinations and thus results in higher travel time.

These kind of patterns are consistently seen in all network variables with the exception of η index and circuitry, C_t . The categorical

Table 3
Mean values of estimated network measures by travel time category.

	Average travel time (minute) to work category			
	Category 1	Category 2	Category 3	Category 4
Independent variables (Unit)	(0,20] min	(20,25] min	(25,30] min	(30, ∞) min
Entropy, $H(X)$	0.77	0.55	0.52	0.39
Percent freeways, F	4.37	2.20	0.73	0.00
Alpha, α	0.17	0.16	0.10	0.10
Arterial Treeness, ϕ_{tree}	0.21	0.26	0.34	0.42
Average Circuitry, C_t	1.41	1.37	1.48	1.44
Eta, η	0.21	0.25	0.42	0.39
Theta, θ	1735.57	1047.80	920.29	476.29
Street Density, ρ_{lm} (km^{-1})	7.50	7.15	3.47	3.46

tabulations show a relation between measures of street network structure and observed travel. The next step was to quantify this relation statistically.

7. Regression models

The estimated models predict the average travel time to work as a function of the independent network variables. Various formulations of the proposed model were analyzed and the results of three select models are presented here. The multimodal (auto, transit) person weighted accessibility to jobs within a 30 min time threshold (A_{30} , T_{30}), the median household income and the size of the minor civil division were used as control variables in all three models.

The first model is a linear regression model with the dependent variable and all independent variables in linear form. The second model is a log-log model similar to the linear regression model but with the dependent variable and all independent variables in natural log form. The third model is a factor analysis model. A review of the residuals from the regression models pointed to the presence of spatial autocorrelation, which violates the assumptions of linear regression, affects standard errors of estimated regression coefficients leading to unreliable results. Spatial regression models are used to address this issue. Spatial regression models are a generalization of linear regression models and explicitly allow and account for spatial effects (Chi and Zhu, 2008; Anselin, 2009).

There are two model approaches that are used in practice to handle spatial dependence in cross sectional data. The first model, namely the spatial error model, views spatial dependence as affecting just the error terms that needs to be eliminated. Consider the standard regression model:

$$Y = X\beta + \epsilon \quad (4)$$

Y denotes the dependent variable, X denotes the explanatory variables, β denotes the regression parameters, and ϵ denotes a vector of normally distributed, homoscedastic, and uncorrelated errors. The spatial error model uses the same formulation but the error term ϵ takes on the following form:

$$\epsilon = \lambda W\epsilon + \mu \quad (5)$$

where λ denotes the spatial autoregressive parameter, W denotes the spatial weights matrix and μ denotes a vector of homoscedastic and uncorrelated errors. The spatial dependence in error terms results in estimates similar to regular regression models but impacts the t-statistics. The estimates from spatial error models can hence be interpreted in the same manner as regular regression models.

The second model is the spatial lag model. Here spatial dependence is given substantive interpretation and is entered into the model with the dependent variable at one location jointly determined by its values at other locations (Anselin and Rey, 1991). The spatial lag model formulation is as follows:

$$Y = \rho WY + X\beta + \epsilon \quad (6)$$

where ρ refers to the spatial autoregressive parameter, WY denotes the spatially lagged dependent variable, and all the other terms are defined as above (Pisati, 2001).

Both spatial lag and spatial error model approaches were tested but based on spatial diagnostic parameters (not presented here), the spatial error model seemed more appropriate. The results from the spatial error models alone are presented in Table 4 for brevity. Additionally, a brief review of the results from the general regression models and the spatial error regression models pointed to consistencies in the magnitude and significance of the independent variables.

Looking at the results presented in Table 4, the entropy variable, $H(X)$, has a significant negative influence on the average travel time. This result is consistent with our hypothesis that an increase in network hierarchy will decrease the travel time. A higher entropy is generally correlated with a more even distribution of links across functional classification. Since many more links are local roads and fewer are high classification like freeways, entropy would be positively correlated with higher class roads and relatively fewer residential streets. Similarly the log-log models confirm that an increase in the average circuitry, C_i , in the network increases the average travel time to work. The other variables of network structure were included in these models but did not show up as statistically significant or were not included due to their correlation with other independent variables.

Now coming to the factor analysis model - factor analysis is a statistical technique that is used to uncover relationships among numerous variables. This technique is used to reduce a large number of correlated independent variables into smaller dimensions, called factors (Stanek, 1995). In this analysis, the principal component factor analysis technique in STATA, a statistical package, was used to reduce the independent network structure variables into factors. The initial factor analysis resulted in a total of eight factors out of which three relevant factors were retained based on best practices in the use of this technique (Costello and Osborne, 2005). The first factor consisted primarily of the α , ϕ_{tree} and C_i variables and was hence referred to as the topology factor. The second factor consisted of the $H(X)$, F and θ variables and was referred to as the hierarchy factor. The third factor consisted of the η and ρ_{lm} variables and hence referred to as the scale factor. A similar approach as described above was used to reduce the person weighted accessibility to jobs by auto and transit ($A_{A,30}$, $A_{T,30}$) to a single relevant accessibility factor.

The rotated factor loadings of the network variables and accessibility variables are provided in Table 5. The three network factors and the accessibility factor were then used as independent variables in the model along with the median household income and the size of the minor civil division to predict the average travel time to work.

The pattern of influence of the factors are based on the underlying variables that make up the factor. For example, the accessibility factor primarily consists of two variables, namely, A_{30} and T_{30} . Based on the hypotheses listed above, both these variables have a

Table 4

Prediction of average travel time to work (minutes) by minor civil division.

Independent variables (Unit)	Spatial error regression models								
	Linear model			Log Model			Factor analysis model		
	Dependent variable - average travel time to work			Dependent variable - ln (average travel time to work)			Dependent variable - average travel time to work		
	Coef.	z	Sig.	Coef.	z	Sig.	Coef.	z	Sig.
Person weighted accessibility to jobs by auto, $A_{A,30}$	−4.393	−8.155	***	−0.147	−4.722	***			
Person weighted accessibility to jobs by transit, $A_{T,30}$	−1.879	−0.991		−0.012	−1.542				
Median Household Income (\$)	1.511	0.191		−0.064	−1.959	*	−2.413	−0.222	
Area of the MCD (km ²)	0.024	3.931	***	0.036	5.453	***	0.041	5.748	***
Entropy, $H(X)$	−2.460	−4.063	***	−0.044	−4.122	***			
Average Circuity, C_t	1.974	1.410		0.158	1.701	*			
Topology Factor ^a							0.089	0.262	
Hierarchy Factor ^b							−0.506	−2.506	**
Scale Factor ^c							0.591	1.575	
Accessibility Factor ^d							−2.435	−5.921	***
Constant	26.154	8.946	***	2.739	23.645	***	23.613	16.719	***
lambda, λ	0.851	5.576	***	−0.232	−0.182		0.734	2.648	***
sigma, σ	2.233	14.859	***	0.055	14.566	***	2.447	15.481	***
Number of observations		145			75			145	
Squared Correlation		0.632			0.722			0.577	
Variance Ratio		0.657			0.714			0.588	

*p < 0.10, ** p < 0.05, *** p < 0.01.

+ Measured at the MCD of residence.

^a Topology Factor uses α (negative loading), ϕ_{tree} and C_t .^b Hierarchy Factor uses θ , F and $H(X)$.^c Scale Factor uses η (negative loading) and ρ_{lm} .^d Accessibility factor uses A_{30} and O_{30} .

* p < 0.10.

** p < 0.05.

*** p < 0.01.

Table 5

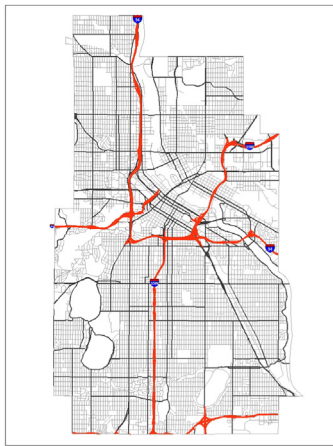
Rotated factor loadings.

Independent Variables	Factor 1 - Topology Factor	Factor 2 - Hierarchy Factor	Factor 3 - Scale Factor	Factor 4 - Accessibility Factor
Entropy, $H(X)$		0.699		
Percent freeways, F		0.801	0.352	
Alpha, α	−0.846		0.402	
Arterial Treeness, ϕ_{tree}	0.839			
Average Circuity, C_t	0.731			
Eta, η			−0.910	
Theta, θ		0.826		
Street Density, ρ_{lm}	−0.457		0.785	
Person weighted accessibility to jobs by auto, $A_{A,30}$				0.899
Person weighted accessibility to jobs by transit, $A_{T,30}$				0.899

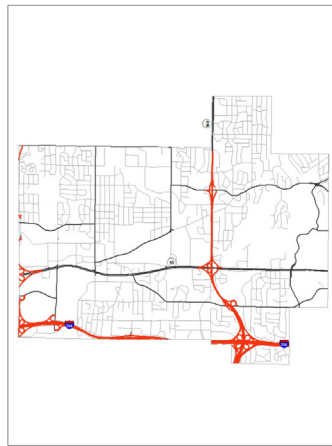
Only loadings with absolute values greater than 0.3 presented.

Loadings represent correlations between independent variables and identified factors.

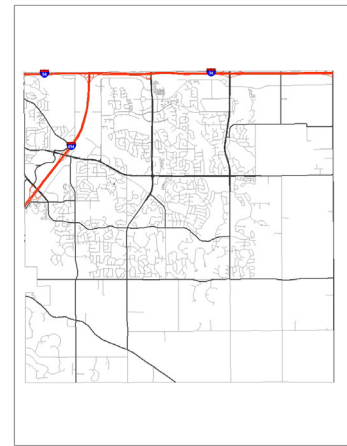
Factor 1 consists primarily of the α , ϕ_{tree} and C_t variables. Referred to as the topology factor.Factor 2 consists of the $H(X)$, F and θ variables. Referred to as the hierarchy factor.Factor 3 consists of the η and ρ_{lm} variables. Referred to as the scale factor.



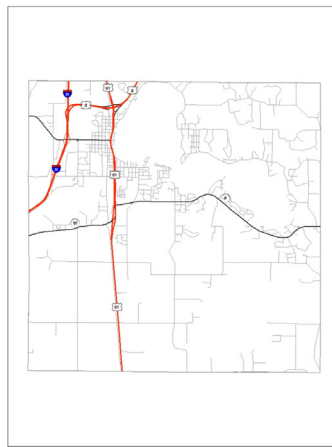
(a) Minneapolis



(b) Golden Valley



(c) Woodbury



(d) Forest Lake

Fig. 4. Street networks of select MCDs from the Twin Cities region.

negative influence on the travel time to work. Hence the accessibility factor would have a negative influence on travel time as well and this is confirmed by the results presented in Table 4. The topology factor has the expected positive influence on the average travel time to work but is statistically insignificant. The hierarchy factor has the expected significant negative influence while the scale factor has a positive influence though insignificant.

The models presented in Table 4 confirm that the network variables influence the travel time to work, after accounting for non-network control variables. The use of different model formulations was to see if the influence of the network variables changes with the model structure. The results show a consistent pattern of influence across the models.

While only select variables were included in the regression models, the results show a relation between certain aspects of network structure and travel. The regression results along with the other analyses (spatial, scatterplot, categorical) point to the importance of considering underlying network structure in understanding travel. On another note, the inclusion of the person weighted accessibility to jobs by auto and transit ($A_{A,30}$, $A_{T,30}$) has a significant influence on model performance, as measured by statistical performance. The importance of using accessibility as an approach to understand the transport system performance has been highlighted in prior research (Levinson, 1998; Levinson and Krizek, 2005; Parthasarathi, 2014). As Levinson et al. (2017) explain “accessibility measures provide a more complete picture of system performance by synthesizing the effects of changes to both transport networks and the destinations they serve”.

8. Discussion

Consider the street networks of four select MCDs from the Twin Cities region, presented in Fig. 4. In a paper on network circuitry, Levinson and El-Geneidy (2009) divided the Twin Cities region into four distinct rings. The first ring (Ring 1) refers to the urban core that includes the cities of Minneapolis and St. Paul. Ring 2 refers to area outside the urban area but within the I-494/I-694 beltway. Ring 3 refers to the area just outside the I-494/I-694 beltway while Ring 4 refers to the outlying areas of the region. Using the same

Table 6
Estimated street network measures and performance for select MCDs.

	Ring 1 Minneapolis	Ring 2 Golden Valley	Ring 3 Woodbury	Ring 4 Forest Lake
Nodes, v	7582	1003	1852	722
Links, e	13,264	1463	2303	927
Entropy, $H(X)$	0.53	0.75	0.43	0.92
Percent freeways, F	2.82	5.78	3.0	3.4
Alpha, α	0.37	0.23	0.12	0.14
Arterial Treeness, ϕ_{tree}	0.03	0.10	0.20	0.31
Average Circuity, C_t	1.16	1.33	1.34	1.47
Eta, η	0.14	0.17	0.19	0.27
Theta, θ	1463.46	1779.39	1103.87	1100
Street Density, ρ_{lm} (km^{-1})	12	9	5	3
Average travel time to work	22.1	19.1	25.4	29.7

approach, the city of Minneapolis is part of the urban core (Ring 1) of the region. The City of Golden Valley lies within Ring 2. The City of Woodbury lies within Ring 3 while the City of Forest Lake belongs to Ring 4. Each of these MCDs have a different connection pattern. Each connection pattern results in a different network structure that in turn results in varying system performance and Table 6 confirms the same.

We can see that the City of Forest Lake has the highest average travel time to work among all four cities. Consider a scenario where the city decides to pursue a goal of reducing the commute time. To reduce the amount of time spent commuting, all else equal, the city could try to build networks that are more direct and better connected, have more higher-level roads, and provide more accessibility, while also implementing land use decisions that increase accessibility.

How can the results presented in this analysis help with this goal? Review the log model presented in Table 4. Since both the dependent and independent variables are log transformed here, the regression coefficient serve as elasticity estimates and measure the percent change in the dependent variable, due to a one percent change in the independent variable. If the entropy, $H(X)$, of the underlying street network in the City of Forest Lake were to increase by 1%, the average travel time to work would reduce by 0.044%. If the city decides to improve accessibility as a way to decrease travel time, a 1% increase in the person weighted accessibility to jobs by auto, A_{30} , results in a 0.147% decrease in the average travel time to work. On the other hand, if the average circuity, C_t , of the underlying street network were to increase by 1%, the travel time in the city would increase 0.158%.

To be clear, time spent commuting might not be the city's only goal, so these changes need to be considered against other policy considerations and implications of good urban form. Further it might not be easy to alter the underlying structure of the transport network, but the results point to how network structure can be a policy tool in design of new facilities or revamping of existing facilities.

9. Conclusions

This paper aimed to analyze the spatial variation of street network structure across Minor Civil Divisions within the Twin Cities metropolitan area and relate it to observed travel. Measures to quantify the street network structure were compiled from existing sources. The spatial variation of network structure across the metropolitan area was analyzed using GIS. Further expanding the spatial analysis, the relationship between specific network measures were analyzed using scatterplots. Finally the relationship between the network measures and observed travel was analyzed using categorical analysis and statistical regression models.

The analyses confirm the inherent variation in the street network structure within a metropolitan area. Specifically the underlying structure of the traditional grid-like networks seen in the central cities of the metropolitan area differs from the structure of the tree-like street networks seen towards the outer edge of the metropolitan area. Further the analyses confirms that some of these differences in underlying street network structure translate into differences in observed travel. The influence of the network measures remain significant, even after controlling for independent non-network variables.

The analysis presented here focuses primarily on the time spent commuting but this research is part of a broader effort to understand and quantify the influence of street network design on observed travel at the individual (Parthasarathi et al., 2012), household (Parthasarathi et al., 2015) and metropolitan level (Parthasarathi, 2014). More importantly, this effort confirmed that the variation in the underlying street network structure can be related to differences in travelers' perception of travel distance and time, which in turn influences actual travel (Parthasarathi, 2011). Other researchers have shown that the design of the transport network has far reaching implications on system performance such as safety; transit ridership; long-term vulnerability; and residential location choice relative to work (Derrible and Kennedy, 2009; Jenelius, 2009; Levinson and El-Geneidy, 2009; Marshall and Garrick, 2010).

Transport networks, specifically street networks, are an integral part of cities. The underlying street network influences the spatial structure of cities and is in turn influenced by the geographic characteristics of cities. There are various considerations at play when designing transport networks. In some cases, geographic constraints (e.g. mountains, water bodies, etc.) limit the type of transport network that can be designed and in other cases land use policies and investment strategies decide the structure of the underlying network. Plans for existing cities can promote accessibility, directness of routes, higher speed limited access routes, and network

connectivity to reduce travel times, and discourage network circuitry and tree-like network structures. Missing links in networks can be completed, and excessive channelization of traffic onto fewer and fewer links (often under the guise of traffic calming) can be resisted. Of course, it must be understood that reducing average travel time is not the only goal of transport systems, but one among many. For instance, diffusing traffic across many routes in a grid network will reduce travel times and congestion on the major streets, but at the cost of increasing traffic levels on residential streets (Appleyard, 1980).

One might argue that this understanding isn't necessarily applicable to the US and other developed countries since the transport networks in these countries have pretty much been built and can be considered to be in a "mature" stage of development. However there are still parts of the US, especially in the south, and the suburban areas of large cities, that are investing heavily in the expansion of their transport networks. Recent media coverage on developing countries highlight the rapid investments in transport systems to meet growing demand, tackle poverty, respond to climate change and sustain economic growth. China's 'One Belt, One Road' initiative is expected to span more than 60 countries and promises massive spending of more than \$1 trillion in new air, rail and road infrastructure (Fallon, 2015; Perlez and Huang, 2017). While not on the scale of China, India is rapidly ramping up its spending to upgrade and modernize its transport infrastructure (Kotoky, 2017). Another factor to consider are the environmental impacts of the transport decisions. The transport sector universally consumes a large share of energy, and given current technologies, remains one of the largest contributors to greenhouse gas (GHG) emissions in the US (United States Environmental Protection Agency, 2017).

The pace and scale of transport investments; the irreversible nature of transport network decisions and the environmental impacts of transport sector highlight the need to better understand the influence of network design. It is critical that network architecture be well-understood and considered in the design of urban form and sustainable environments. Clear consideration of network design allows us to learn from the past and better plan for future transport systems.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.tra.2018.09.008>.

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